MD-88 Audio Companion: Lessons Learned

Getting the as-ordered board (Y15, fab_v7) working on the bench. Bring-up dates 2026-11-12. Master notes: `BRINGUP_TODO.md`, `audio_companion/RESPIN_CHECKLIST.md`.

1. Result

The board now runs from a LiPo cell with the 12 V boost, passes audio to the headphones and Lemo, and the mic sidetone is clear. Getting there needed **two hardware reworks** (relay bridge, mic bias resistor), one **battery hookup**, and several firmware fixes. All of the reworks are bench workarounds for design errors that must be fixed in the next revision.

2. Hardware reworks on this board

#	Where	What was done	Why
1	Headphone relay `HEADPHONE_BYPASS-1` (G6K-2F-Y)	Pads **3 and 4** bridged (bodge wire / solder blob) and pads **5 and 6** bridged.	Coil polarity is reversed on the board, so the relay never operates. The bridge forces the amp onto PHONES/Lemo. Removes the fail-safe: bench use only.
2	Mic bias: `JP-MIC-BIAS-1` and Lemo J-LEMO-1	BIAS solder jumper left **open**. A **980 Ω** resistor soldered between Lemo **pin 6 (+12 V)** and **pin 4 (MIC_RAW)**, sleeved over pin 5 (AGND).	On-board bias (3V3 via 2.2 kΩ, 2.69 V under load) was weak and noisy. 12 V rail via 980 Ω gives 6.2 V and about 6.6 mA at the mic, with clear sidetone.
3	Battery	LiPo on **J-BAT-2**. A PH2.0 jumper from CrowPanel **J3** to **J-BAT-1** ties VBAT so the CrowPanel TP4059 charger charges the same cell. Polarity verified with a meter. ANR power switch (IO8) enables the boost: 12.7 V on Lemo pin 6.	The companion has no charger of its own. The ANR rail sits at about 3.3 V when off (MT3608 leakage path), so a dim ANR LED is normal.

Bench only. With the relay unoperated, the mic line (MIC_RAW) reaches the aircraft mic jack, so the 12 V bias would also appear there. Do not connect to an aircraft with these reworks fitted.

3. What went wrong and what we learned

Problem	Root cause	Lesson
No audio on PHONES/Lemo	Relay coil connected backwards: pin 1 to the drive transistor, pin 8 to	For every polarized part (relays, diodes, caps, ICs) check the pin function against the manufacturer datasheet, not the library symbol. Add a +/– mark to the symbol.

(always the raw aircraft path)	3V3. The G6K-2F-Y is polarized (pin 1 = +, pin 8 = –).	
Weak, noisy mic	Bias from 3V3 through 2.2 kΩ; Wi-Fi ripple on 3V3 couples into the audio.	Bias an electret mic from a filtered higher rail, never the digital supply. Provide a resistor footprint, not a solder jumper, so the value can be tuned. Bench-test the real headset on the bench supply before layout.
Mic gain looked dead	ES8388 ADC digital volume powers up at –96 dB.	Read the codec reset values. The firmware now sets 0 dB.
Mic channel and clipping surprises	Mono mic on the left channel only; ES8388 L/R slot order varied between starts; firmware gains were set for the weak bias.	Provide an Auto channel select; return gains to 0–6 dB preamp and 0–12 dB sidetone with proper bias.
BTN_IN button cannot be read	Routed to PCA9534 U3 pin 13, which is the INT output.	Check each GPIO-expander pin direction before assigning a signal. Move to pin 12 (SPARE5).
USB debugging lost	IO19/IO20 are the ESP32-S3 native USB pins and were used for MCLK and BYPASS_EN.	Keep native USB pins free. Flash over the CH340 or the Wi-Fi hotspot (AudioBox-Update).
Battery path risks	No reverse protection, fuse, switch or charger on the companion. The 12 V rail never fully turns off.	Add protection and a switch; either tap the CrowPanel VBAT or add a charger.
First order cancelled (Y14)	Mounting holes and encoder did not clear the CrowPanel; panel connectors up to 8.9 mm tall; import duties pushed the order near $400.	Measure the real panel first (calipers), model the stack in 3D with collision checks, and include duties when comparing assembly options. Do not assume gaps.
Board shorted on the table	Exposed underside on a conductive surface.	Use an insulating mat for bench work.
Docs and files scattered	Notes and KiCad files lived in different folders and drifted.	One master location: the AudioBox repo, `audio_companion/RESPIN_CHECKLIST.md`.

4. Bench procedure that worked (in order)

1. Meter the battery polarity and the jumper cable end to end; use a protected single cell.
2. Charge from the CrowPanel with USB, companion unplugged; confirm the charge LED.
3. Add the J3 to J-BAT-1 jumper and the cell on J-BAT-2. Switch ANR power on and confirm 12.7 V at Lemo pin 6.
4. Open the BIAS jumper and add the 980 Ω resistor between Lemo pins 6 and 4; check there is no short to AGND (pin 5) first.
5. Plug in the headset: expect about 6.2 V at the mic and clear sidetone in both ears.
6. For headphones, bridge HEADPHONE_BYPASS-1 pads 3–4 and 5–6. Do not bridge the mic relay.

5. Fix list for the next revision

- Swap the coil nets on both relays (pin 1 to 3V3, pin 8 to K1_COIL_DRIVE).
- Mic bias from +12 V through about 1 kΩ with RC filter, gated off in bypass; resistor footprint instead of the solder jumper.
- Move BTN_IN to U3 pin 12; move MCLK and BYPASS_EN off IO19/IO20.
- Battery reverse protection, fuse, switch; gate the ANR LED or add a load switch so off means off.
- Test points: mic bias node, codec LIN1/RIN1, +12 V.
- Rotate the zip-tie slots 90 degrees.

6. Pre-fab checklist (apply before the next order)

- Check every polarized part pin-by-pin against its datasheet.
- Check every GPIO and expander pin for direction and reserved functions (USB, strapping).
- Check every power rail for loads, off-state behavior, and reverse/short protection.
- Prototype analog bias and gain on the bench before committing the layout.
- Verify mechanical fit with measured parts and the 3D stack model; include duties in the cost.
- Run an independent netlist review before ordering.